

WhatsApp Automation App — UI/UX Prototype

A complete desktop-first HTML/CSS/JS prototype suite for a Python WhatsApp automation application.
Designed for eventual PySide6/PyQt implementation.

File Tree

```
whatsapp_app_ui/  
├─ index.html           # Dashboard (shell + stats + quick launch + recent activity)  
├─ profiles.html        # Profile management (cards + empty/loading/error states)  
├─ contacts.html        # Contact lists (table with inline edit + bulk actions)  
├─ wa_contacts.html     # WhatsApp directory sync (synced contacts table)  
├─ groups.html          # WhatsApp groups (group list + member extraction)  
├─ templates.html       # Message templates (list + editor + preview)  
├─ send_messages.html   # Send workflow (3-column: recipients + composer + preview)  
├─ schedule.html        # Scheduled jobs (status table + edit modal)  
├─ logs.html            # Send history (log table + filter + pagination)  
├─ styles.css           # Global design system (~33KB, all components)  
└─ app.js               # Interactions (modals, toasts, tabs, command palette)
```

How to Run

- 1. Download the folder whatsapp_app_ui/
- 2. Open any .html file in a modern browser (Chrome, Firefox, Edge, Safari)
- 3. Navigate between pages via the left sidebar
- 4. Try interactions: buttons, checkboxes, modals, command palette (Ctrl+K)

No build step, no server, no dependencies.

Design Architecture

Layout

```
+-----+  
| Sidebar | Topbar (title + search + actions) |  
|         +-----+  
| (nav)   | Main Content Area                 |  
|         | (scrollable, max-width 1200px)    |  
|         +-----+  
| Profile | [Optional Right Panel]             |  
| mini    | (preview / help / details / logs)  |  
+-----+
```

Design Tokens

- **Backgrounds:** Base #0b0f14, Elevated #111820, Surface #161f2b
- **Accent:** WhatsApp green #22c55e with glow effects
- **Border radius:** 6px–12px for cards, 9999px for pills
- **Spacing:** 4px base unit (4, 8, 12, 16, 20, 24, 32, 40)
- **Typography:** System sans + SF Mono for code/numbers
- **Shadows:** Layered for depth (sm/md/lg)

Component Library (in styles.css)

Component	CSS Class	Notes
Button (primary)	.btn.btn-primary	Green glow on hover
Button (danger)	.btn.btn-danger	Red tint, destructive actions
Button (ghost)	.btn.btn-ghost	Low emphasis
Button group	.btn-group	Segmented actions
Card	.card	Surface + border + radius
Table	.data-table	Hover states, sticky header
Checkbox	.checkbox	Custom styled, JS toggle
Chip/Tag	.chip.*	Color-coded: green/blue/amber/red/gray
Status pill	.status.*	Dot + label: pending/dispatched/success/error/idle
Modal	.modal-overlay + .modal	Centered, animated, overlay click to close
Toast	.toast-container + .toast	Bottom-right, auto-dismiss
Command Palette	.command-palette	Ctrl+K trigger, search + shortcuts
Empty state	.empty-state	Icon + title + description + CTA
Loading state	.loading-state + .spinner	Centered spinner
Error state	.error-state + .error-icon	Red icon + message
Dropzone	.dropzone	Drag-over highlight
Progress bar	.progress-bar + .progress-fill	Animated width
Segmented control	.segmented + .segmented-item	Toggle groups
Tabs	.tabs + .tab	Underline active indicator
Form input	.form-input / .form-select / .form-textarea	Focus ring

Table 1 – continued

Component	CSS Class	Notes
Attachment chip	<code>.attachment-chip</code>	File name + remove button
Variable chip	<code>.var-chip</code>	Template variable insert
Preview bubble	<code>.preview-bubble</code>	Chat-style message preview
Stats card	<code>.stat-card</code>	Big number + label + delta
Launch card	<code>.launch-card</code>	Dashboard quick-nav cards
Recipient item	<code>.recipient-item</code>	Send page selectable row
KBD shortcut	<code>.kbd-shortcut</code>	Keyboard hint styling

Page-by-Page Specs

index.html — Dashboard

- **Stats row:** 4 cards (profiles, contacts, pending jobs, messages sent)
- **Quick launch:** 8 cards in responsive grid, each links to a module
- **Recent activity:** Last 4 log entries inline

profiles.html — Profiles

- **Toolbar:** New Profile, Refresh
- **Profile cards:** Avatar (gradient initials), name, phone, status badge, actions
- **States shown:** Normal, Empty (demo at bottom), Loading (demo at bottom), Error (session expired)
- **Modals:** Add Profile, Delete Confirmation (with cascade warning)

contacts.html — Contacts & Lists

- **Toolbar:** List selector dropdown, Add/Rename/Delete/Columns, Import CSV
- **Bulk actions bar:** Appears when rows selected (green tint)
- **Table:** Inline editable cells, checkbox selection, hover actions
- **Modal:** New List (name + custom fields)

wa_contacts.html — WhatsApp Contacts

- **Toolbar:** Open WhatsApp, Load from WhatsApp, Export CSV, Clear Saved
- **Table:** Name, phone (mono font), WhatsApp status, last synced, send action
- **Bulk send bar:** Bottom bar when recipients selected
- **Modal:** Clear confirmation (explains non-destructive scope)

groups.html — WhatsApp Groups

- **Toolbar:** Open WhatsApp, Load Groups, Add Members, Clear Saved
- **Table:** Group name, member count chip, last synced, send action

- **Modal:** Add Members (explains extraction process + warning), Clear Groups

templates.html — Templates

- **2-column layout:** Template list (left) + Editor (right)
- **List:** 5 templates, selected state with green left border
- **Editor:** Name, message body, variable chips (click to insert), emoji grid, live preview bubble
- **Modal:** Delete Template

send_messages.html — Send Messages

- **3-column layout:** Recipients (1fr) + Composer (360px) + Preview/Status (320px)
- **Recipients:** Segmented toggle (Lists/Directory/Groups), search, selectable rows with avatars, select-all
- **Composer:** Profile selector, template selector, message textarea, variable chips, dropzone with 2 attachment chips, attachment-only checkbox
- **Actions:** Send Now (primary, disabled until selection) + Schedule (secondary)
- **Preview:** Chat bubble with rendered template + attachments list
- **Status:** Profile, recipient count, message length, attachments, progress bar
- **Shortcuts:** Ctrl+Enter, Ctrl+Shift+S, Ctrl+A
- **Modal:** Schedule (date + time + timezone + summary)

schedule.html — Schedule

- **Toolbar:** Refresh, Edit Selected, Delete Selected, status filter segmented
- **Table:** Run At (mono), Status (pill), Target, Recipients (chip), Message, Items, Last Error
- **6 demo rows:** Pending, Dispatched, Completed, Failed
- **Hint:** Double-click to edit pending
- **Modal:** Edit Job (current job info, new datetime picker, message edit)

logs.html — Logs

- **Toolbar:** Refresh, status filter (All/Sent/Failed), Clear All Logs
- **Table:** Time (mono), Profile (chip), Recipient, Message, Status, Retry action
- **7 demo rows:** Mix of sent/failed across 3 profiles
- **Pagination:** Showing X of Y, Prev/Next buttons
- **Modal:** Clear confirmation (shows total count, warns permanence)

Interactions (in app.js)

Interaction	Trigger	Behavior
Sidebar active	Page load	Highlights current page nav item
Command Palette	Ctrl+K	Opens overlay + search input
Modal open	data-modal click or openModal(id)	Fade in + scale animation

Table 2 – continued

Interaction	Trigger	Behavior
Modal close	Escape, overlay click, X button	Fade out
Toast	<code>showToast(msg, type)</code>	Bottom-right, 4s auto-dismiss
Checkbox	Click	Toggle checked state + row selection
Select All	Master checkbox click	Toggles all rows in table
Tab switch	Click <code>.tab</code>	Activates tab + shows matching panel
Segmented	Click <code>.segmented-item</code>	Activates one item
File dropzone	Drag over / drop	Visual feedback + toast
Recipient select	Click row	Toggles checkbox + updates count + enables Send Now
Variable insert	Click <code>.var-chip</code>	Inserts at cursor in textarea
Search trigger	Click <code>.search-trigger</code>	Opens command palette
Demo toasts	<code>data-toast</code> attribute	Shows toast on button click
Demo modals	<code>data-modal</code> attribute	Opens modal on button click

PySide6/PyQt Mapping

HTML/CSS	Qt Widget	Notes
<code>.app-shell grid</code>	<code>QMainWindow + QSplitter</code>	Use splitters for resizable panels
<code>.sidebar</code>	<code>QDockWidget</code> or custom <code>QWidget</code>	Fixed width, scrollable nav
<code>.topbar</code>	<code>QToolBar</code>	Add to main window top
<code>.card</code>	<code>QFrame + setFrameShape(QFrame.StyledPanel)</code>	Add shadow via stylesheet
<code>.data-table</code>	<code>QTableView + QAbstractTableModel</code>	For large datasets; <code>QTableWidget</code> for small
<code>.form-input</code>	<code>QLineEdit</code>	Set placeholder, focus ring via QSS
<code>.form-textarea</code>	<code>QTextEdit</code>	Plain text mode for composer
<code>.form-select</code>	<code>QComboBox</code>	Custom popup styling
<code>.btn-primary</code>	<code>QPushButton</code>	QSS for green bg, glow on hover
<code>.modal-overlay</code>	<code>QDialog</code> with <code>Qt.FramelessWindowHint</code>	Darken parent via <code>QGraphicsOpacityEffect</code>
<code>.toast-container</code>	Custom <code>QWidget</code> with <code>QPropertyAnimation</code>	Slide up + fade in

Table 3 – continued

HTML/CSS	Qt Widget	Notes
.command-palette	QDialog + QLineEdit + QListView	Filter model on text change
.checkbox	QCheckBox	Custom QSS for styled checkbox
.segmented	QButtonGroup + QPushButton	Exclusive checkable buttons
.tabs	QTabWidget	Or custom QStackedWidget + buttons
.dropzone	QWidget with dragEnter/drop events	setAcceptDrops(True)
.progress-bar	QProgressBar	Custom QSS for rounded + gradient
.preview-bubble	QTextEdit (read-only) or QLabel	Rich text for formatting
.recipient-item	QListWidgetItem or custom QWidget	With checkbox + labels
.launch-card	QPushButton or QToolButton	Clickable card with icon + text
.stat-card	QFrame	Big QLabel + small QLabel
.status pill	QLabel with QSS	Rounded + colored dot
.chip	QLabel with QSS	Inline tag styling
.attachment-chip	QFrame + QLabel + QToolButton	File name + remove X
.var-chip	QPushButton	Click to insert text
.empty-state	QWidget + centered layout	Icon + title + description + CTA
.loading-state	QWidget + QMovie (spinner GIF)	Or custom QProgressIndicator
.error-state	QWidget + red styling	Error icon + message + retry button

Recommended Qt Layouts

- **Dashboard:** QGridLayout for stats + QFlowLayout for launch cards
- **Profiles:** QGridLayout or QListView with custom delegate
- **Contacts table:** QTableView with QStyledItemDelegate for inline editing
- **Send page:** QSplitter (horizontal) with 3 panels
- **Templates:** QSplitter (vertical) — list top, editor bottom, or horizontal split
- **Schedule/Logs:** QVBoxLayout — toolbar + QTableView

Quick Wins for Rollout

1. **Implement the shell first** — sidebar + topbar + main area grid. All pages plug into this.
2. **Use QSS (Qt Stylesheets)** — map CSS variables to QSS. The `styles.css` is your QSS reference.
3. **Start with Dashboard** — lowest risk, proves navigation and layout system.

4. Use **QStackedWidget** for page switching instead of multiple windows. Sidebar buttons switch index.
 5. **Command palette last** — it's polish. Implement basic menu actions first.
 6. **Toast system early** — users need feedback. A simple **QTimer**-based popup is enough.
 7. **Modal dialogs reusable** — one **QDialog** subclass, configure title/body/buttons per use.
 8. **Table models for data** — **QAbstractTableModel** + **QSortFilterProxyModel** for search/filter.
 9. **Profile status as enum** — Ready/Idle/Error/Loading. Drive UI colors from state.
 10. **Keyboard shortcuts** — **QShortcut** for Ctrl+K (palette), Ctrl+Enter (send), Escape (close modal).
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UX Principles Applied

- **Top-down + left-to-right:** Global context (topbar) → Navigation (sidebar) → Primary workflow (main) → Details (right panel)
- **Minimal clicks:** Bulk actions, inline editing, keyboard shortcuts, command palette
- **Clear states:** Every page shows empty, loading, error, and success states
- **Confirmation + undo:** Destructive actions have modals with explicit consequences
- **Visual hierarchy:** Stats → Cards → Tables → Details, with consistent spacing and color coding
- **Desktop-first:** Fixed sidebar, wide tables, multi-column layouts, hover-reveal actions
- **Progressive disclosure:** Advanced options (schedule, custom fields) behind modals or toggles
- **Feedback loops:** Toasts for actions, progress bars for sends, status pills for jobs